Sepehr Hajebi – CV

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Contact

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Appointments

- **University of Waterloo**, Combinatorics and Optimization — 2024 – now
  Postdoctoral Scholar (Advisor: Sophie Spirkl)
- **Princeton University**, Department of Mathematics — 2024 – never
  Junior Faculty (Instructor)
  (Not taken up due to visa delays)

Education

- **University of Waterloo**, PhD in Combinatorics and Optimization — 2020 – 2024
  Thesis: *Foreshadowing the Grid Theorem for Induced Subgraphs*
  Advisor: Sophie Spirkl
- **Isfahan University of Technology**, BSc and MSc in Mathematics — 2012 – 2019

Awards

- **Graduate Research Excellence Award, 2025** (2500 CAD)
  Faculty of Mathematics, University of Waterloo
- **Mathematics Doctoral Prize: 1st place, 2025** (1500 CAD)
  Faculty of Mathematics, University of Waterloo
- **Governor General's Gold Medal, Finalist Designation, 2025**
  University of Waterloo
- **Sinclair Graduate Scholarship, 2023-2024** (4000 CAD)
  Combinatorics and Optimization, University of Waterloo
- **Outstanding TA Award, 2022**
  Combinatorics and Optimization, University of Waterloo
- **Math competitions:**
  International Mathematics Competition for University Students: Third Prizes, 2015 and 2016
  Korean Mathematical Society Contest: Third Prize, 2016
  Iranian Mathematical Society Competition: Second and Third Prizes, 2015 and 2016
- **Elite Student Recognition, 2016 and 2018**
  Isfahan University of Technology

Talks

- *Polynomial pathwidth*
  Barbados Graph Theory Workshop, Bellairs Research Institute, Holetown, Barbados, 2026

- *Complete bipartite induced minors (and treewidth)*
  Willaim Tutte Colloquium, University of Waterloo, 2025
- *Local and global structure of non-basic obstructions to bounded treewidth*
  CanaDAM (invited at the Structural Graph Theory Minisymposium), University of Ottawa, 2025.
- *"Underview" of excluded induced subgraphs for bounded pathwidth*
  Barbados Graph Theory Workshop, Bellairs Research Institute, Holetown, Barbados, 2025
- *The pathwidth theorem for induced subgraphs*
  IBS discrete math seminar, Daejeon, South Korea, 2025
- *The pathwidth theorem for induced subgraphs*
  Graphs and Matroids Seminar, University of Waterloo, 2024
- *Anticomplete sets of large treewidth*
  BIRS Workshop on New Perspectives in Colouring and Structure, 2024
- *Survey on induced subgraphs of graphs with large treewidth*
  Barbados Graph Theory Workshop, Bellairs Research Institute, Holetown, Barbados, 2024
- *Treewidth, Erdős-Pósa and induced subgraphs* [virtual]
  New York Combinatorics Seminar, 2024
- *Several Gyárfás-Sumner-type results for treewidth*
  Graphs and Matroids Seminar, University of Waterloo, 2023
- *Hitting all maximum stable sets in P_5-free graphs*
  Graphs and Matroids Seminar, University of Waterloo, 2023
- *Forests in even-hole-free graphs of large treewidth*
  Barbados Graph Theory Workshop, Bellairs Research Institute, Holetown, Barbados, 2022
- *Holes, hubs, and bounded treewidth* [virtual]
  IBS Discrete Math Seminar, 2022
- *Bounded treewidth in hereditary graph classes*
  Graphs and Matroids Seminar, University of Waterloo, 2022
- *Bounded treewidth in hereditary graph classes*
  Seymour was 70 two years ago, ENS de Lyon, France, 2022

Teaching

- **Courses taught at Waterloo**
  - CO 380 Mathematical Discovery and Invention (Spring 2026)
  - MATH 135 Algebra for Honours Mathematics (Winter 2026)
  - MATH 116 Calculus I for Engineering (Fall 2025)
- **Courses TA'ed as PhD student at Waterloo**
  - CO 342 Graph Theory (Spring 2023)
  - MATH 138 Calculus II for Honors Mathematics (Winter 2023)
  - MATH 600 Mathematical Software (Fall 2022)
  - CO 456 Game Theory (Fall 2022 and 2023)

- CO 380 Mathematical Discovery and Invention (Spring 2022)
- MATH 239 Introduction to Combinatorics (Winter 2022)
- CO 255 Advanced Optimization (Winter 2022)
- CO 250 Introduction to Optimization (Fall 2021 and 2023, Winter 2023 and 2024, Spring 2024)
- CO 442/642 Graph Theory (Fall 2021)
- CO 351 Network-Flow Theory (Spring 2021)

- **Courses TA'ed as BSc/MSc student at Isfahan University of Technology**
  - Computational Complexity (2019)
  - Elements of Matrices and Linear Algebra (2018)
  - Applied Linear Algebra for Engineering (2018)
  - Graph Theory (2014, 2016, 2017)
  - Elements of Combinatorics (2014, 2016, 2017)

Service

- **Supervision/Mentorship**
  - Graduate students:
    - Xinyue Fan (MSc, 2024 – , joint with Sophie Spirkl)
  - Directed Reading Program (DRP):
    - Spring 2024 project: *Non-constructive methods in combinatorics*
      Mentees: Anna Henderson and Gioia De Leonardis
    - Fall 2023 project: *Introduction to graph minor theory*
      Mentees: Xinyue Fan and Lyncy Li
  - Undergraduate Research Assistant Program (URA):
    - Spring 2023 project: *Maximum transitive set in H-free tournaments*
      Mentee: Yun Xing
- **Organizing**
  - *Themes and trends in structural graph theory* – Organizer
    Workshop at the American Institute of Mathematics (AIM), to be held in 2027
- **Refereeing for journals and conference proceedings:** IMRN, Combinatorica, JCTb, Eur. J. Combin., SIDMA, J. Graph Theory, Innov. Graph Theory, Electron. J. Combin., SODA, EuroComb, Workshop on Graphs.